

Payton Henry

Portfolio Website: <https://peh-portfolio.netlify.app/>

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Education

California Baptist University, Riverside, CA – Expected Graduation: June 2027

Bachelor's in Computer Science & concentration in Machine Learning and AI

Current Cumulative GPA 3.89 | President's List (2 terms), Provost's List (5 terms)

Technical Skills and Developer Tools

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, Apex, SOQL/SOSL, HTML, CSS

Systems & Platforms: Linux (Ubuntu Server, Arch), Docker, TAK (FreeTAKServer, PyTAK), GCP, Firebase, Salesforce (Agentforce, Experience Cloud, CLI)

Frameworks & Tools: Next.js, React, React Native/Expo, Lightning Web Components, Gemini API, Git/GitHub, Jira, Claude Code, VS Code, IntelliJ, AutoCAD

Experience

Salesforce Developer with CAiMLL

May 2025–Present | California Baptist University, Riverside, CA & NavalX TechBridge, Norco, CA

- Design advanced Salesforce solutions using Apex, SOQL, Triggers, Flows, Lightning Web Components, and AI agents through Agentforce to automate and enhance user workflows
- Develop AI agents using natural language processing and machine learning to use unique data querying and accelerate the process of report writing
- Build responsive Lightning Web Components using HTML, CSS, and JavaScript to create seamless user interfaces integrated within Salesforce
- Designed 10+ custom Salesforce data models as part of the engineering team, creating the schema foundation for a production system now tracking 35,000+ records
- Contribute to a 10-person engineering team delivering in weekly sprints, using GitHub version control, Jira project management, and manifest-driven deployments with Salesforce CLI

Projects

TAK Live Aircraft Tracking Demo | CAiMLL

September 2026 - Present

- Built a Python pipeline using PyTAK that converts live ADS-B data from the adsb.fi API into Cursor-on-Target (CoT) messages, streaming CBU aviation program aircraft and Riverside City Buses that stop on campus as part of their route onto a TAK map in real time using only public data
- Deployed FreeTAKServer in Docker on a self-managed Ubuntu Server 24.04 machine to serve as the TAK backend
- Pivoted from the official TAK Server to open-source FreeTAKServer when tak.gov access requirements blocked the official downloads, keeping the full pipeline working end to end

Formula Fly | Connectome-Driven Reinforcement Learning | In Progress

September 2026 - Present

- A simulated fruit fly sees a race track through a visual system wired from the real FlyWire connectome and learns to drive an F1 car with reinforcement learning, testing whether biological wiring helps (Python, PyTorch, MuJoCo, flyvis)
- Built the MuJoCo Silverstone racing simulator and a Ferrari SF70H model calibrated from Assetto Corsa data
- Added lap timing, live telemetry, and gamepad input, and connected the simulator to the team's flyvis harness

Ember | Graph-Based Context Retrieval for AI Coding Agents | Senior Capstone | In Progress

Fall 2026 - Present

- Builds a knowledge graph of a team's decisions, tickets, people, and code from GitHub, Jira, and Slack, served to AI coding agents through an MCP server to replace static context files (Next.js, TypeScript, Python, Docker, Kubernetes)
- Built the Next.js web app with a D3 graph explorer and a dashboard that flags where GitHub, Jira, and Slack disagree
- Built the Python Slack ingestion service with HMAC request verification, replay protection, and rate-limit backoff

SnapDose | AI-Powered Diabetes Tracking App | Project Manager

January 2026 - May 2026

- A React Native/Expo app that combines live Dexcom glucose data, meal tracking, and insulin dosing in one dashboard, estimating carbs from meal photos with Gemini (Firebase, GCP Cloud Functions, Spring Boot)
- Led the project as Project Manager and built the Gemini meal-photo pipeline and all insulin dosing and calculations
- Set up Firebase Authentication and Firestore security rules so each user's diabetes data stays private

Volunteer and Leadership Experience

October 2023 | Bournes Technology Center, Riverside, CA

STEP Convention

- Lead a showcase of an engineering project about plastic deformation and tensile strength for youth who were interested in engineering to inspire them to pursue their interests in the STEM field.